

Electromagnetic Induction

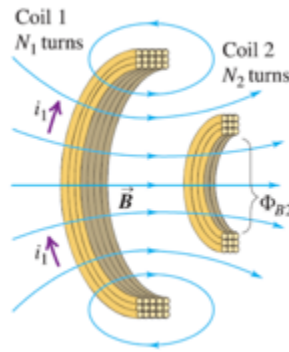
Induced electromotive forces (emf) provide the basis behind electric motors and electrical power generation. Faraday's Law states that a **changing** magnetic flux, through a coil of N turns, induces an emf $\mathcal{E}$ given by

$$\mathcal{E} = -N \frac{d}{dt} \Phi_B,$$

where the **magnetic flux** Φ_B is defined as

$$\Phi_B = \int \vec{B} \cdot d\vec{A} = \cos(\theta) \int B dA.$$

An induced emf will only be present when the magnetic flux **changes** with time. That is, when the strength of the magnetic field B , the area A , or the angle between $\vec{B}$ and $\vec{A}$ varies with time. The faster the flux changes, the larger $\mathcal{E}$ will be. Additionally, $\mathcal{E}$ can be amplified by increasing the number of turns N in a coil.



One method of creating a changing magnetic field is by putting a time varying current through a coil. This coil is then called the **primary coil** and is labeled Coil 1 in the diagram above. Coil 1 carries the current $i_1(t)$ which produces the magnetic field $B_1(t)$. This magnetic field then creates a magnetic flux $\Phi_{B2}(t)$ through the **secondary coil**, labeled Coil 2 above. This flux $\Phi_{B2}(t)$, is a function of the geometry and separation of the two coils and can be a difficult quantity to calculate since $B_1(t)$ is, in general, not uniform. However, since $\Phi_{B2}(t)$ is proportional to $B_1(t)$ and therefore also proportional to $i_1(t)$, we can write that the emf $\mathcal{E}_2$ induced in the secondary coil is

$$\mathcal{E}_2 = -M \frac{d}{dt} i_1.$$

Here, M is a constant of proportionality called the **mutual inductance**. M has the unit henry (H) and is purely a function of the geometry, separation and orientation of the two coils.

Goals of this Lab

1. Induce emf in coils by hand
2. Understand how induced emf is created

Lab Materials

Coil of Wire

An induced emf will be produced in a coil of $N = 3200$ turns. The coil is wound on a form that contains banana plug connectors.



Neodymium Bar Magnet

Induced emfs in the coil will be produced moving a strong magnet inside the coil.



Light Emitting Diode (LED)

LEDs will be lit with an induced current.



Differential Voltage Probe

The emfs will be recorded with a computer-controlled voltmeter.



Logger Pro

Logger Pro will be used for data analysis.



Logger Pro

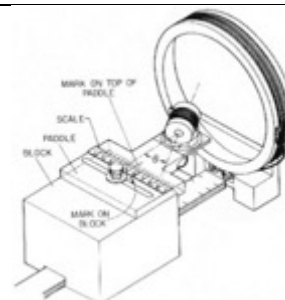
Function Generator

The function generator will be used to produce a time varying current in a coil.



Primary Coil and Secondary Coil

Time varying currents (and thus, time varying magnetic fields) will be produced in a large coil. Induced emfs will be produced in a secondary coil that can be positioned at various points in the changing magnetic field.



Hantek Digital Oscilloscope

Time varying emfs will be measured with a computer-controlled oscilloscope.



PC Oscilloscope Software

The voltages across Coil 1 and 2 will be monitored on the computer.



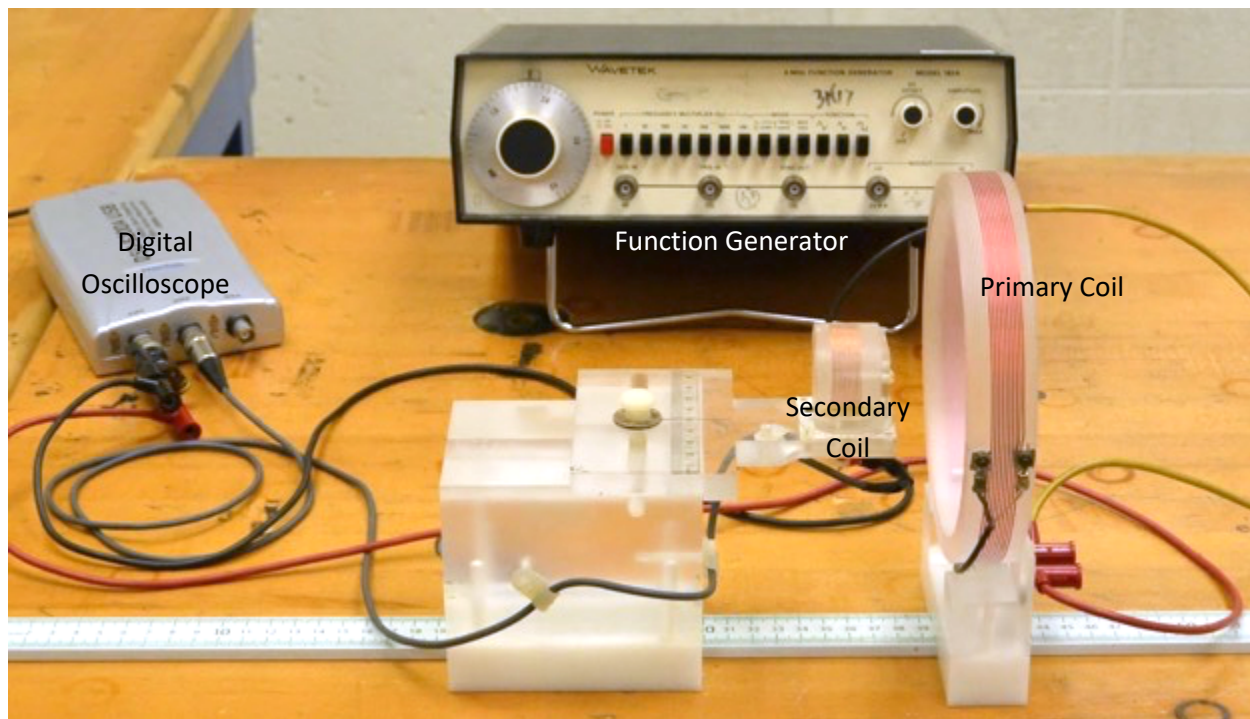
DSO-5200A

Overview

In this lab, you will begin by inducing an emf, literally by hand. By moving a magnet through a coil of wire, the changing magnetic flux in the wire will induce an emf, which will be measured by Logger Pro. Also attached to the coil are two LED's, wired in opposing directions. Since LED's will only light when current is supplied in one direction, this will provide a visual guide to the direction of the emf.

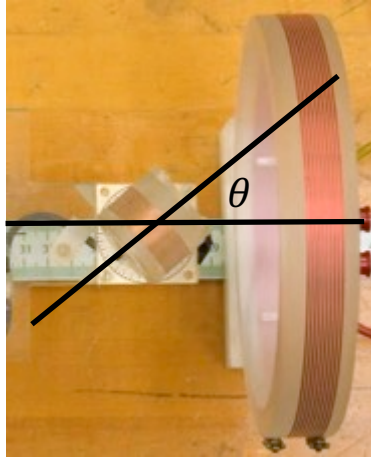


In the second part of the lab, you will switch to using an automated magnetic field. A function generator creates a time-dependent current $i_1(t)$ through a large coil, which we will call the primary coil. This changing current then produces the time-dependent magnetic field $B_1(t)$. We can then observe the induced emf $\mathcal{E}_2(t)$ in the smaller secondary coil created by that field.



The function generator can produce a time varying voltage that varies sinusoidally ($\sim$), as square waves ($\square$) or as triangle waves ($\triangle$). Each of these waveforms will produce a time-varying magnetic field in the primary coil, which will in turn produce a time-varying magnetic flux $\Phi_{B2}(t)$ through the second coil,

and hence an induced emf. This secondary coil can also be rotated by the angle θ relative to the plane of the primary.



The magnetic field at the **center** of the primary coil, of radius R and N turns and carrying a time-varying current $i_1(t)$, is given by

$$B_1(t) = \frac{\mu_0 N}{2R} i_1(t).$$

Therefore, (approximating B as a constant across the face of the secondary coil) the magnetic flux through a secondary coil of radius r and n turns is

$$\Phi_{B2}(t) = B_1(t) A \cos(\theta) = \frac{\mu_0 N}{2R} i_1(t) n \pi r^2 \cos(\theta).$$

θ is the angle between the axes of the coils. The induced emf $\mathcal{E}_2$ in the secondary coil is then

$$\mathcal{E}_2(t) = \frac{-d}{dt} \Phi_{B2} = \frac{-\mu_0 n N \pi r^2}{2R} \cos(\theta) \frac{d}{dt} i_1 = -M(\theta) \frac{d}{dt} i_1.$$

Here $M(\theta) = \frac{\mu_0 n N \pi r^2}{2R} \cos(\theta)$ is called the mutual inductance. It has units of $V \cdot \frac{s}{A} = H$ (Henry) and depends on the geometry of the coils.

For a **sinusoidal** waveform, the current $i_1(t)$ in the primary coil is given by

$$i_1(t) = i_{1,\max} \cos(2\pi f_1 t).$$

Here, $i_{1,\max}$ and f_1 are the current amplitude and frequency, respectively. Therefore,

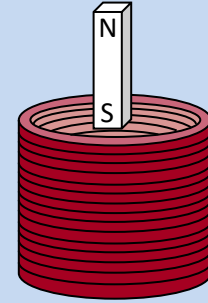
$$\mathcal{E}_2(t) = -M(\theta) \frac{d}{dt} i_1(t) = -M(\theta) 2\pi f_1 i_{1,\max} \sin(2\pi f_1 t) = -\mathcal{E}_{2,\max} \sin(2\pi f_1 t).$$

Therefore, $\mathcal{E}_{2,\max} = M(\theta) 2\pi f_1 i_{1,\max}$ is the amplitude of the induced emf.

Pre-Lab Exercise

A bar magnet is moved relative to a coil as shown in the diagram. Which direction (clockwise or anti-clockwise) will the induced current flow if:

1. The magnet is moving into the coil
2. The magnet is moving out of the coil
3. The coil rotates clockwise until it makes a 45° angle with the magnet



Lab Procedure

Observe induced emf with a coil and bar magnets

A changing magnetic field will produce an emf in a coil of wire. Here you will induce an emf by moving a magnet relative to a coil. Remember that the two LED's are wired in opposing directions so you can tell the direction of the current by observing which LED lights up.

1. Orient the coil so that you have access to both ends of the coil and can watch the LED's.
2. Check that the LED plugs and voltage probes are attached as shown in the photo.
3. Light emitting diodes (LEDs) will emit light when supplied by a few milliWatts (mW). Try to light both LED's.
4. Start Logger Pro and start collecting data and move the bar magnet in and out of the coil of wire. Observe the effect of the speed at which you move the magnet. Try moving the magnet across the face of the coil without entering it.



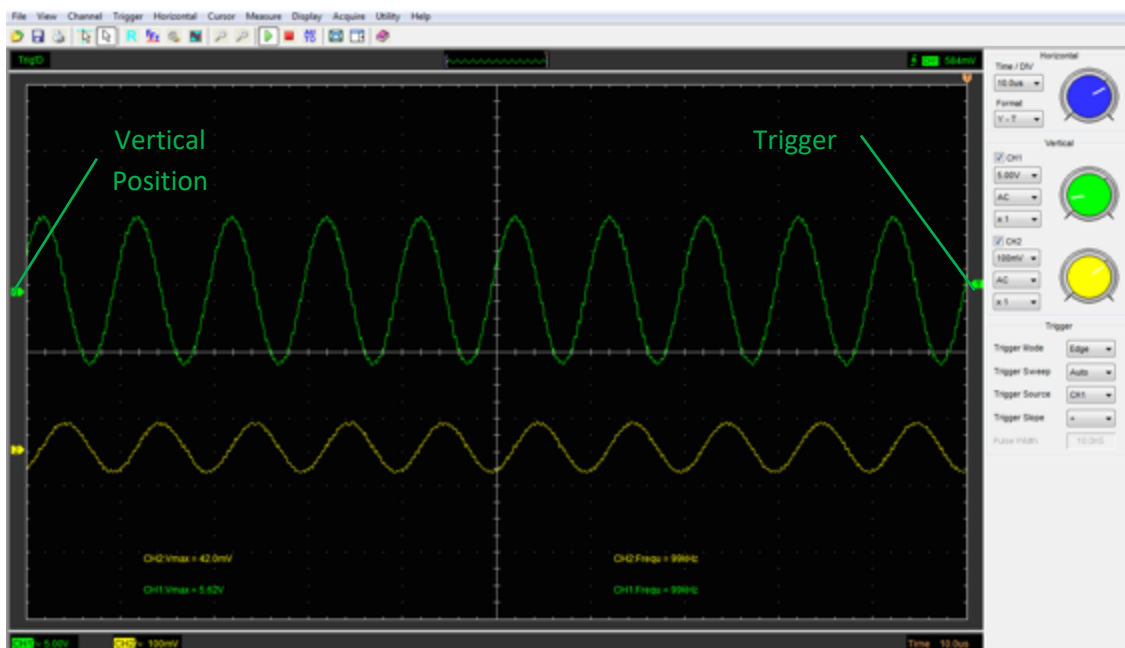
Questions

1. How could you generate a larger emf with the bar magnet?
2. You should have observed that the voltage peaked in one direction for moving the magnet in and peaked in the other direction for moving the magnet out. Why is this?
3. By inducing positive and negative voltage peaks in your coil, determine the magnet's north and south pole. Follow the direction of the windings to determine the direction of the induced current. Describe how you determined the poles of your magnet. Check your prediction with a compass.
4. From the data you collected, what was the maximum induced current, $i_{\max}$, in your coil? What determines the value of the current? **Hint:** The coil has a resistance of 150Ω .

Observe AC waveforms using the oscilloscope

You can observe time varying voltages using an oscilloscope. The oscilloscopes we will use are computer-controlled and can be accessed via the DSO-5200A USB icon  on the computer Desktop.

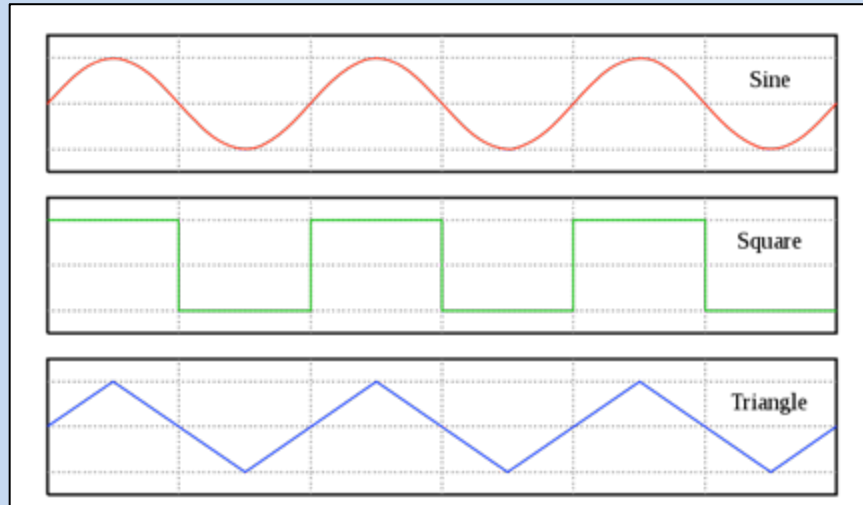
1. Open DSO-5200A USB icon  on the computer Desktop
2. Turn on the function generator and set the amplitude to the maximum value. Dial the frequency to 100 kHz.
3. Observe the waveform on the oscilloscope. Below is an example of what you should see. You may have to tune several of the parameters on the right-hand side of the display in order to observe the wave.
 - i. Horizontal axis (Time/DIV): The vertical dotted line divisions are separated by $10\mu\text{s}$ in the example below. The higher the frequency of the input wave, the smaller the Time/DIV should be. Adjust these divisions in the Time/DIV box on the right-hand side.
 - ii. Vertical axis (Volts/DIV): The horizontal dotted line divisions are separated by 5.00V for CH1 and 100mV for CH2 in the example below. The larger the amplitude of the input waveform, the larger the Volts/DIV should be. Adjust these divisions in the box below CH1 on the right-hand side.
 - iii. Vertical Position: The waveform can be shifted up or down for better viewing by clicking and dragging on the arrow on the left-hand side of the screen
 - iv. Trigger: The oscilloscope will sweep across the input waveform based on when the input signal crosses a certain threshold. In order to observe a stable periodic signal, the trigger should be lined up with part of the input waveform as shown below.



- The oscilloscope can calculate the peak value and frequency of your waveforms. From the top menus, select "Measure" → "Source" → "CH1", then "Measure" → "Vertical" → "Maximum" and "Measure" → "Horizontal" → "Frequency". Repeat for CH2.

Pre-Lab Exercise

- The graphs below show the current in Coil 1 as a function of time. Make a sketch of the induced current as a function of time. Remember that an emf, and thus current, is related to the derivative of the magnetic flux $\frac{d\Phi_{B2}}{dt}$.



Investigate induced emf from a time varying magnetic field

- Arrange the primary and secondary coil such that they are both aligned on the meter stick and the center of the coils are separated by 1-3 cm.
- Observe the signals from the primary and secondary coil. One channel is the voltage across the primary coil $\mathcal{E}_1(t)$ and the other channel is the induced emf in the secondary coil $\mathcal{E}_2(t)$. The Time/DIV scale is common to both signals. You can vary the Volts/DIV scale independently in order to observe both signals on the one screen. You can move the vertical positions of both signals by moving the color-coded arrow on the left side of the screen.
- Observe the induced emf $\mathcal{E}_2(t)$ in the secondary coil for each input waveform, sine (⤿), square (⏏) and triangle waves (⤶).

Questions

- Describe the shape of the waveform in the secondary coil for a sine, square and triangle wave in the primary coil. How does the signal in the secondary coil relate to the one in the primary coil?
- How does the slope of the input waveform relate to the induced emf?
- Are the induced emf and applied voltage waveforms in phase? That is, do the peaks and valleys of the two waveforms line up? Why or why not?

4. Once you have oriented yourself with the oscilloscope, set the input waveform to a sinusoidal waveform. Using the frequency knob, vary the frequency f_1 within the range 50 kHz to 150 kHz. Using the amplitude knob, vary the amplitude of the primary coil $\mathcal{E}_{1,\max}$.
5. Using data from the oscilloscope, in a separate graphing program, like Logger Pro, make plots of the following:
 - 1) $\mathcal{E}_{1,\max}$ (primary coil amplitude) vs. $\mathcal{E}_{2,\max}$ (secondary coil amplitude)
 - 2) f_1 (primary coil frequency) vs. f_2 (secondary coil frequency)
 - 3) f_1 vs. $\mathcal{E}_{2,\max}$
 - 4) $\cos \theta$ (orientation of secondary coil) vs. $\mathcal{E}_{2,\max}$

Be sure to isolate each variable while creating your plots. For example, while changing the primary amplitude with the amplitude knob, be careful to not change the coil position or orientation.

Questions

8. Submit the plots you created in step 4. How do the variables $\mathcal{E}_{1,\max}$ and f_1 relate to the induced voltage $\mathcal{E}_{2,\max}$ and f_2 ?
9. From your plot of f_1 vs. $\mathcal{E}_{2,\max}$, what is the value for M ? **Hint:** The coil has a resistance of 0.5Ω .
10. How does the secondary coil signal change with θ ? What is the emf induced in the secondary coil when its face is oriented perpendicular ($\theta = 90^\circ$) to the face of the primary coil? How is this consistent with Faraday's Law as stated at the top of the lab?
11. List all relevant quantitative results and conclusions. Check that all graphs are included and your report is in the correct order. How do you feel about your performance in this lab?

Feedback

Do you have any comments, complaints, or suggestions about this lab?
Feel free to email our Lab Manager directly at Pharnish@physics.upenn.edu.